YOLO

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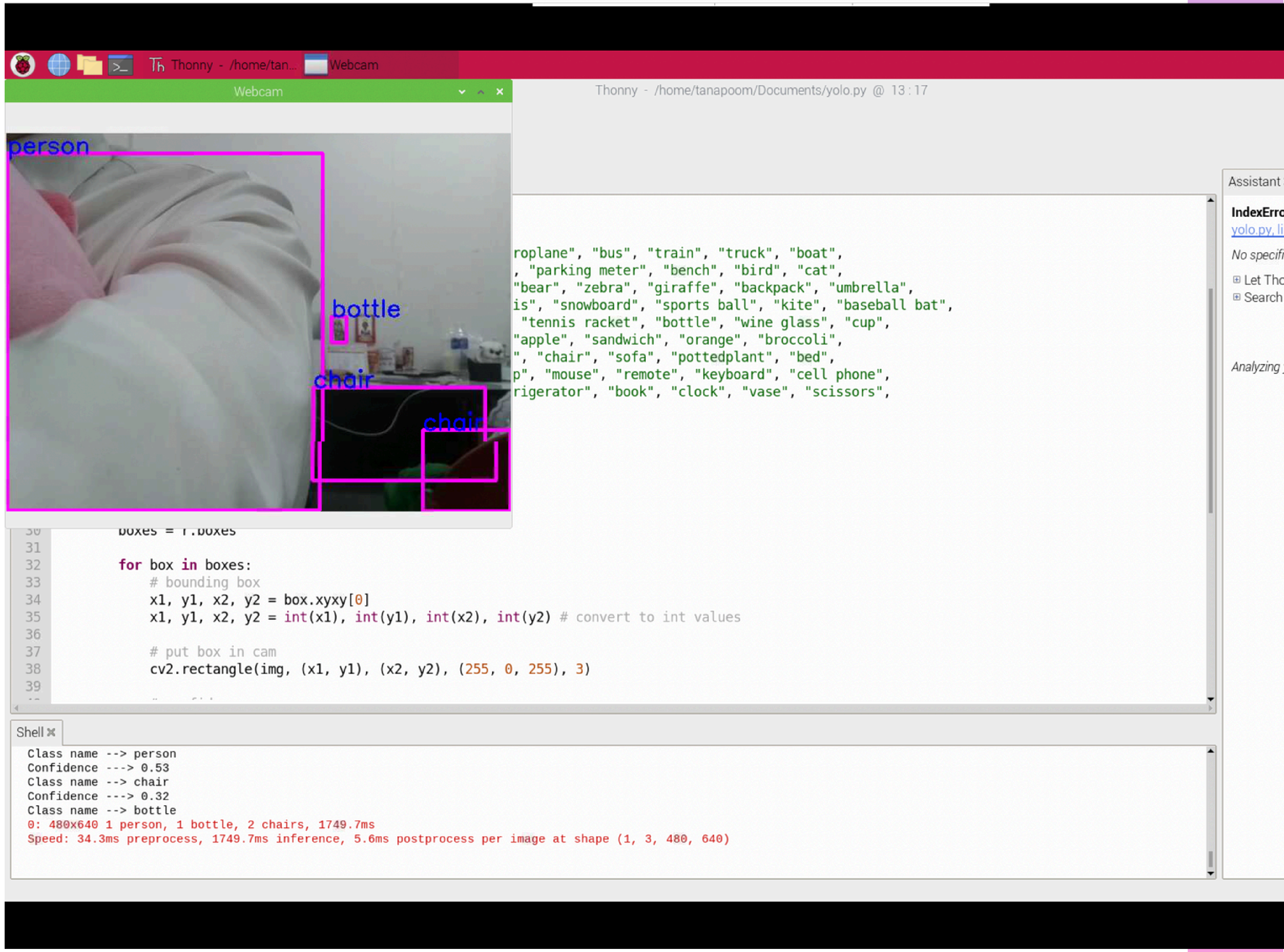
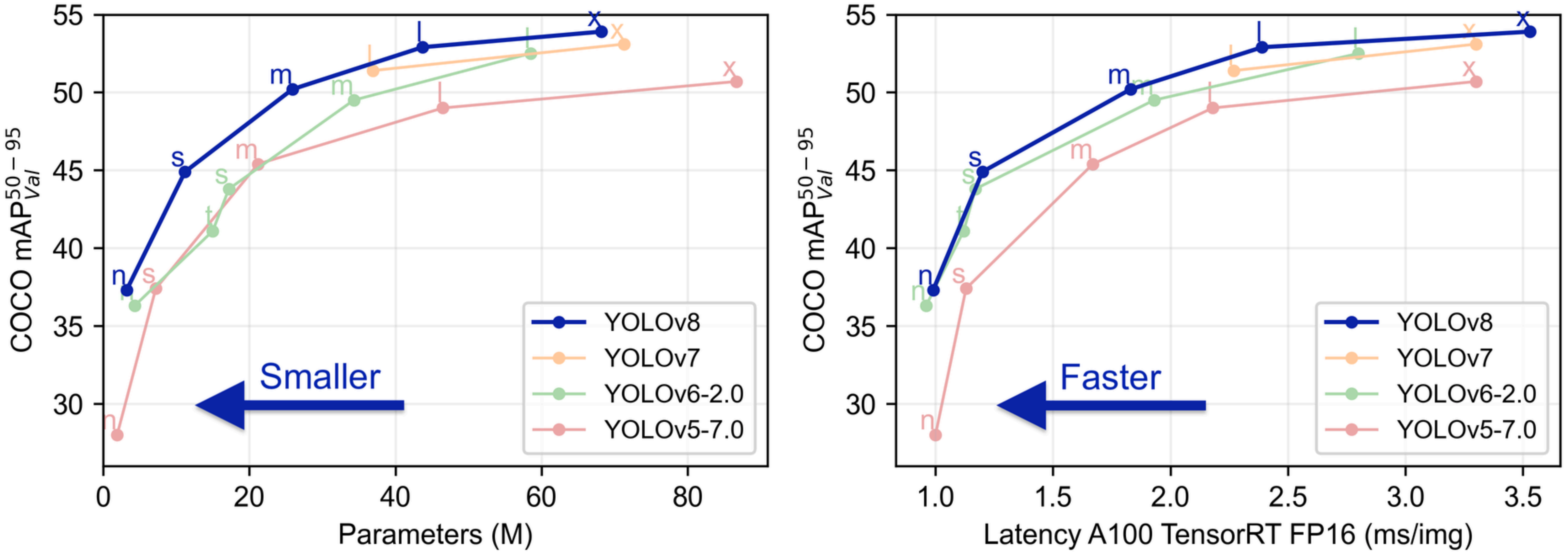
YOLO หรือ You Only Look Once คือ Realtime Object Detection Model ที่มีความโดดเด่นเรื่องความเร็วและความถูกต้อง



YOLO

- ปัจจุบัน YOLO มาถึง version ที่ 11 แล้วโดยมีหลายแบบจำลองให้เลือกใช้ (และมีโครงสร้างที่ไม่เหมือนกัน)



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Model	size (pixels)	mAP^{val} 50-95	Speed CPU ONNX (ms)	Speed A100 TensorRT (ms)	params (M)	FLOPs (B)
YOLOv8n	640	37.3	80.4	0.99	3.2	8.7
YOLOv8s	640	44.9	128.4	1.20	11.2	28.6
YOLOv8m	640	50.2	234.7	1.83	25.9	78.9
YOLOv8l	640	52.9	375.2	2.39	43.7	165.2
YOLOv8x	640	53.9	479.1	3.53	68.2	257.8

การลง library

ใช้คำสั่ง
pip install ultralytics

requirements in a Python>=3.8 environment with PyTorch>=1.8.

code

```
from ultralytics import YOLO
import cv2
import math
import time
# start webcam
cap = cv2.VideoCapture(0)
cap.set(3, 640)
cap.set(4, 480)
# model
model = YOLO("yolo-Weights/yolov8x.pt")
#model.setInputParams(size=(640,480), scals = 1/255)

# object classes
classNames = ["person", "bicycle", "car", "motorbike", "aeroplane", "bus", "train", "truck", "boat",
              "traffic light", "fire hydrant", "stop sign", "parking meter", "bench", "bird", "cat",
              "dog", "horse", "sheep", "cow", "elephant", "bear", "zebra", "giraffe", "backpack", "umbrella",
              "handbag", "tie", "suitcase", "frisbee", "skis", "snowboard", "sports ball", "kite", "baseball bat",
              "baseball glove", "skateboard", "surfboard", "tennis racket", "bottle", "wine glass", "cup",
              "fork", "knife", "spoon", "bowl", "banana", "apple", "sandwich", "orange", "broccoli",
              "carrot", "hot dog", "pizza", "donut", "cake", "chair", "sofa", "pottedplant", "bed",
              "diningtable", "toilet", "tvmonitor", "laptop", "mouse", "remote", "keyboard", "cell phone",
              "microwave", "oven", "toaster", "sink", "refrigerator", "book", "clock", "vase", "scissors",
              "teddy bear", "hair drier", "toothbrush"
              ]
```

```python

while True:
    success, img = cap.read()
    results = model(img, stream=True)
    for r in results:
        boxes = r.boxes

        for box in boxes:
            # bounding box
            x1, y1, x2, y2 = box.xyxy[0]
            x1, y1, x2, y2 = int(x1), int(y1), int(x2), int(y2) # convert to int values

            # put box in cam
            cv2.rectangle(img, (x1, y1), (x2, y2), (255, 0, 255), 3)

            # confidence
            confidence = math.ceil((box.conf[0]*100))/100
            print("Confidence --->",confidence)

            # class name
            cls = int(box.cls[0])
            print("Class name -->", classNames[cls])

            # object details
            org = [x1, y1]
            font = cv2.FONT_HERSHEY_SIMPLEX
            fontScale = 1
            color = (255, 0, 0)
            thickness = 2

            cv2.putText(img, classNames[cls], org, font, fontScale, color, thickness)

    cv2.imshow('Webcam', img)
    if cv2.waitKey(1) == ord('q'):
        break

cap.release()
cv2.destroyAllWindows()
```